

SERAT M. SAAD

Ph.D. Student in Astronomy, The Ohio State University

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RESEARCH INTERESTS

Dynamics: binary stars, radial migration, tests of Modified Newtonian Dynamics (MOND), an alternative to dark matter

Stellar Spectroscopy: chemical abundances, metal-rich stars, star spots, young stellar objects

AI for Astronomy: LLM agents, agent coordination, automation of astrophysical research

EDUCATION

Ph.D. in Astrophysics (in progress), The Ohio State University 2025-Present
Advisors: Prof. Yuan-Sen Ting and Prof. Krzysztof (Kris) Stanek

A.B. in Physics & Applied Mathematics, Vanderbilt University 2021-2025
Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra
Advisor: Prof. Keivan Stassun

HONORS & AWARDS

University Distinguished Fellowship, The Ohio State University 2028-2029
graduate-school fellowship supporting doctoral study

University Fellowship, The Ohio State University 2025-2027
graduate-school fellowship supporting doctoral study

Hult Prize, National Finalist 2026
for a student venture expanding access to personalized education

Larry R. Cathey Award, Vanderbilt University 2025
annual departmental award for the top graduating student studying astronomy

Highest Honors in Astronomy, Vanderbilt University 2025
for distinction in the undergraduate honors thesis and its defense

International Full-ride Student Grant, Vanderbilt University 2021-2025
full four-year scholarship covering all costs of undergraduate study

Dean's List, Vanderbilt University 2021-2025
eight semesters of academic distinction

Buchanan Fellowship, Vanderbilt University 2022
supported a data-science project applying NLP to British periodical collections

Silver Medal, International Olympiad on Astronomy and Astrophysics 2020
represented Bangladesh at IOAA 2019 and the Global e-Competition (GeCAA) 2020

PUBLICATION SUMMARY

12 papers, 7 as first author; 5 published in refereed journals, 6 under review, and 1 preprint.
102 citations from 99 citing papers ([NASA ADS Library](#)).

FIRST AUTHOR PUBLICATIONS AND PREPRINTS

7. **Serat M. Saad** & Yuan-Sen Ting (2026)
“More than Half of Recent Astronomy Papers are Written with Language-Model Assistance”
[*arXiv:2609.10664*](#)
6. **Serat M. Saad** & Yuan-Sen Ting (2026)
“Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra”
[*The Open Journal of Astrophysics, submitted*](#)
5. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek, et al. (2026)
“PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy”
[*The Open Journal of Astrophysics, submitted*](#)
4. **Serat M. Saad** & Yuan-Sen Ting (2026)
“No Gravitational Anomaly in Wide Binaries from Forward Modeling of 3D Orbits”
[*The Open Journal of Astrophysics, 9*](#)
3. **Serat M. Saad** & Yuan-Sen Ting (2026)
“High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND)”
[*The Open Journal of Astrophysics, submitted*](#)
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun, et al. (2025)
“ABYSS III: Observing Accretion Activity in Young Stars through Empirical Veiling Measurements”
[*The Astronomical Journal, 170, 87*](#)
1. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun, et al. (2024)
“ABYSS II: Identification of Young Stars in Optical SDSS Spectra and Their Properties”
[*The Astronomical Journal, 167, 125*](#)

CO-AUTHOR PUBLICATIONS AND PREPRINTS

5. Qinghui Sun, **Serat M. Saad**, Yuan-Sen Ting, et al. (2026)
“Planets Around Solar Twins/Analogues (PASTA) III: Chemical Clock Relations in Planet-Hosting Solar Analogues”
[*Astronomy & Astrophysics, submitted*](#)
4. SDSS Collaboration, including **Serat M. Saad** (2026)
“The Twentieth Data Release of the Sloan Digital Sky Surveys”
[*The Astrophysical Journal Supplement Series, submitted*](#)
3. Zimo Cheng, Sharon X. Wang, Fan Liu, . . . **Serat M. Saad**, et al. (2026)
“A Bayesian Search for Planet Engulfment Signatures in Solar Analogues”
[*AAS Journals, submitted*](#)
2. SDSS Collaboration, including **Serat M. Saad** (2026)
“The Nineteenth Data Release of the Sloan Digital Sky Survey”
[*The Astrophysical Journal Supplement Series, 285, 9*](#)
1. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu, Yuting Shen (2026)
“Egent: An Autonomous Agent for Equivalent Width Measurement”
[*The Open Journal of Astrophysics, 9*](#)

TELESCOPE TIME ALLOCATIONS

Principal Investigator

Large Binocular Telescope, PEPsi, 20 hours · PANTERA: ultra metal-rich stars	2026B
Large Binocular Telescope, PEPsi, 20 hours · PASTA: 36 planet-hosting stars	2026A
Large Binocular Telescope, PEPsi, 28 hours · PASTA: 43 planet-hosting stars	2025B

Co-Investigator

Automated Planet Finder, Levy, 40 hours · metal-rich stars (PI: D. M. Rowan)	2026B
Chandra X-ray Observatory, Cycle 26 · Pink Mergers (PI: Prof. Adi Foord)	2024

INVITED TALKS AND CONFERENCE PRESENTATIONS

LLM Agents Seminar, NASA AI/ML STIG <i>Hands-on session II: agentic coding and research tools</i>	2026
Monday Science Seminar, University of Wisconsin–Madison <i>Title to be announced</i>	2026
AAS 245 Winter Meeting, American Astronomical Society <i>An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)</i>	2025
Cool Stars 22, Cambridge <i>Using Low Resolution Optical Spectra to Identify Young Stars and Their Properties</i>	2024
IEEE Quantum Computing and Engineering (QCE) <i>Developing Workforce in Quantum Industry: The Wond'ry Quantum Studio</i>	2023

TEACHING & MENTORING

Teaching and grading in astronomy, statistics, data science, and programming, alongside mentoring olympiad students in Bangladesh and graduate-school applicants from underrepresented backgrounds.

Mentor, Project SHORT <i>free admissions mentoring for underrepresented applicants to graduate and medical programs</i>	2025
Mentor, Bangladesh Olympiad on Astronomy and Astrophysics <i>national competition that selects and trains Bangladesh's team for the international olympiad (IOAA)</i>	2021-2024
TA, DS 2100: Statistics for Data Science, Vanderbilt University	Spring 2025
TA, ASTR 1010: Introductory Astronomy, Vanderbilt University	Spring 2024
Grader, DS 1100: Introduction to Data Science, Vanderbilt University	Fall 2023
Instructor, Quantum Computing for Everyone, The Wond'ry Vanderbilt	Fall 2023
Grader, CS 1100: Introduction to Problem Solving with Python, Vanderbilt University	2022-2023
Tutor, Vanderbilt Student Athletics	2022

OTHER EXPERIENCES

Building an educational AI initiative, communicating new research through writing, and several years training Bangladesh's astronomy olympiad team.

Co-founder, PorAI <i>educational AI initiative bringing personalized, tutor-like instruction to students and schools</i>	2026-Present
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Author and Editor, Astrobites	2026-Present
Journal Referee (A&A)	2026-
Academic Team Member, Bangladesh Olympiad on Astronomy and Astrophysics	2021-2024
Co-lead, The Wond'ry Quantum Studio	2022-2023
<i>student-led quantum computing program at Vanderbilt's innovation center</i>	
Team Leader, International Olympiad on Astronomy and Astrophysics	2022
<i>led Bangladesh's national team and served as a question moderator at the international round</i>	

NON-REFEREED PUBLICATIONS

6. **Serat M. Saad** (2026)
 “‘Total Eclipse of the Heart’ in Astronomy”
[Astrobites, August](#)
5. **Serat M. Saad** (2026)
 “Where Have All the Star Couples Gone?”
[Astrobites, August](#)
4. **Serat M. Saad** (2026)
 “Cosmic Cannibalism: When Stars Eat Their Planets”
[Astrobites, May](#)
3. **Serat M. Saad** (2026)
 “Wormholes Might Be More Real Than We Thought”
[Astrobites, April](#)
2. **Serat M. Saad** (2026)
 “A Wandering Supermassive Black Hole Eating a Star”
[Astrobites, March](#)
1. **Serat M. Saad**
 “Pixels and Persuasion”
As Seen on TV, digital exhibition, Vanderbilt University Libraries

REFERENCES

Krzysztof (Kris) Stanek, The Ohio State University <i>University Distinguished Scholar; Guggenheim Fellow (2022)</i>	Ph.D. Advisor, stanek.32@osu.edu
Yuan-Sen Ting, The Ohio State University <i>Inaugural Chair, NASA AI/ML Science and Technology Interest Group; Kavli Scholar (2026)</i>	Ph.D. Advisor, ting.74@osu.edu
Keivan G. Stassun, Vanderbilt University <i>National Medal of Science (2024); MacArthur Fellow (2024)</i>	Undergraduate Advisor, keivan.stassun@vanderbilt.edu
Marina Kounkel, University of North Florida	Research Advisor, marina.kounkel@unf.edu
Jessie C. Runnoe, Vanderbilt University	Research Advisor, jessie.c.runnoe@vanderbilt.edu